



PIN Diode Based FIRE SENSOR

D. MOHAN KUMAR

Here is an ultra-sensitive fire sensor that activates an alarm when it detects fire. Thermistor based fire alarms have a drawback; the alarm turns on only if the fire heats the thermistor in close vicinity. In this circuit, a sensitive PIN diode is used as a fire sensor for a longer-range fire detection.

It detects visible light and infrared (IR) in the range of 430nm - 1100nm. So visible light and IR from the fire can easily activate the sensor to trigger the alarm. It also detects sparks in the mains wiring and, if these persist, it gives a warning alarm.

It is an ideal protective device for showrooms, lockers, record rooms and so on. Author's prototype is shown in Fig. 1.

PIN diode BPW34 (Fig. 2) is used in the circuit as light and IR sensor. BPW34 is a 2-pin photodiode with anode (A) and cathode (K). The anode end can easily be identified from the top-view flat surface of the photodiode. A small solder point to which a thin wire is connected is the anode and the other one is the cathode terminal.

BPW34 is a tiny PIN photodiode or mini solar cell with radiant sensitive surface that generates 350mV DC open-circuit voltage when exposed to 900nm light. It is sensitive to natural sunlight and also to light from fire. So it is ideal for use as a light sensor.

BPW34 photodiode can be used in zero-bias as well as reverse-bias states. Its resistance decreases when light falls on it.

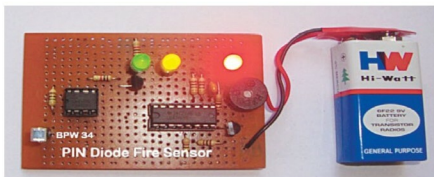


Fig. 1: Author's prototype

Circuit and working

Circuit diagram of the PIN diode based fire sensor is shown in Fig. 3. It is built around 9V battery, PIN diode BPW34 (D1), op-amp CA3140 (IC1), counter CD4060 (IC2), transistors BC547 (T1 and T2), a piezo buzzer (PZ1) and a few other components.

In the circuit, PIN photodiode BPW34 is connected to the inverting and non-inverting inputs of op-amp IC1 in reverse-biased mode to feed photo current into the input of op-amp. CA3140 is a 4.5MHz Bi-MOS op-amp with MOSFET inputs and bipolar output.

Gate-protected MOSFET (PMOS) transistors in the input circuit provide very high input impedance, typically around 1.5T ohms. The IC requires very low input current, as low as 10pA, to change output state to high or low.

In the circuit, IC1 is used as a transimpedance amplifier to act as a current-to-voltage converter. IC1 amplifies and converts the photo current generated in the PIN diode to the corresponding voltage in its output. The non-inverting input is

PARTS LIST

Semiconductors:

- IC1 - CA3140 op-amp
- IC2 - CD4060 counter
- T1, T2 - BC547 npn transistor
- LED1-LED3 - 5mm LED
- D1 - BPW34 PIN photodiode

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R5, R6 - 1-mega-ohm
- R2, R3 - 1-kilo-ohm
- R4, R7, R8 - 100-ohm

Capacitor:

- C1 - 0.22 μ F ceramic disk

Miscellaneous:

- BATT.1 - 9V battery
- PZ1 - Piezo buzzer



Fig. 2: BPW34 PIN photodiode

connected to the ground and anode of photodiode, while the inverting input gets photo current from the PIN diode.